

cwltool

System	No Container	Singularity
LUMI	ERROR (3)	INVALID (29)
MareNostrum 5	OK (8)	ERROR (3)
Laptop	OK (3)	INVALID (4)
Cloud	OK (5)	INVALID (5)

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System	No Container	Singularity
LUMI	ERROR (124)	INVALID (113)
MareNostrum 5	ERROR (18)	ERROR (10)
Laptop	OK (11)	INVALID (14)
Cloud	OK (15)	INVALID (20)

Table 1: Simple MPI workflow using **baseCommand: mpirun**. Execution time is shown in seconds. “INVALID” indicates runs where the MPI job was launched via Singularity instead of the host MPI launcher.

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System	No Container	Singularity
LUMI	OK (4)	ERROR (4)
MareNostrum 5	OK (3)	ERROR (2)
Laptop	OK (3)	ERROR (3)
Cloud	OK (4)	ERROR (6)

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System	No Container	Singularity
LUMI	OK (14)	ERROR (20)
MareNostrum 5	ERROR (8)	ERROR (9)
Laptop	OK (11)	ERROR (14)
Cloud	OK (14)	ERROR (19)

Table 2: Simple MPI workflow using **code:MPIRequirement**. Execution time is shown in seconds.